

# Rockchip\_Developer\_Guide\_Android12+\_HD MI\_IN\_Bridge\_CN

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前言

本文档是基于RK356X/RK3588/RK3562 Android12平台，使用HDMI转MIPI-CSI2/DVP桥接芯片开发HDMI IN/DPIN功能的帮助文档。

概述

HDMIIN功能可以通过桥接芯片，将HDMI信号转换成MIPI信号接收，RK3588/RK356X等平台MIPI接口可以支持该功能开发。本文档主要以RK3588 Android12平台为例，介绍通过HDMI/DP转MIPI-CSI桥接芯片开发HDMI IN功能的方法。

产品版本

| 芯片名称                 | 内核版本       | Android版本    |
|----------------------|------------|--------------|
| RK3588/RK3562/RK356X | Linux 5.10 | Android12/13 |

读者对象

本文档（本指南）主要适用于以下工程师：

技术支持工程师

软件开发工程师

修订记录

| 版本号    | 作者  | 修改日期       | 修改说明     |
|--------|-----|------------|----------|
| V1.0.0 | 范建威 | 2023-9-27  | 初始版本     |
| V1.0.1 | 范建威 | 2023-12-27 | 修复一些错误描述 |
|        |     |            |          |

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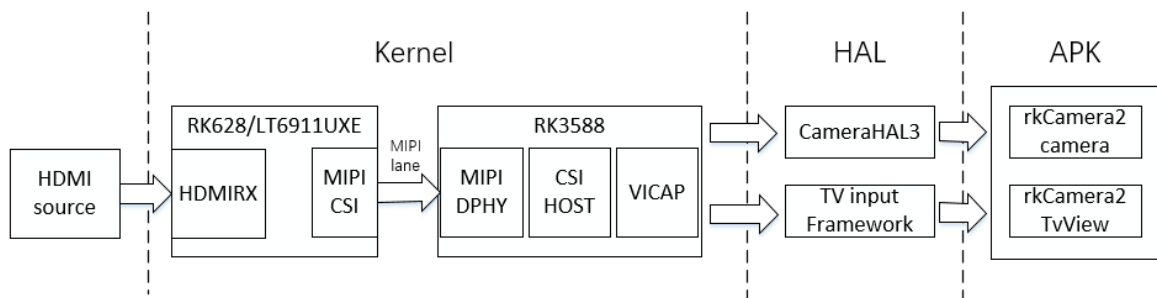
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## 1. HDMI IN功能概述

### 1.1 HDMI TO MIPI-CSI2 功能框图



根据应用场景需求，HDMI TO MIPI-CSI2需要搭配一颗转接芯片（RK628D/LT6911UXE），可适配TV框架（后TIF同）或是Camera框架，适配TIF框架图像传输延时更低，适配Camera框架可以使用标准Camera API，更方便录像、对接后端算法等应用功能开发。

## 1.2 RK SOC接收能力

HDMI/DP-IN桥接芯片方式，类似于SOC Camera设备，接入数量以及接收规格受限于MIPI PHY和DVP接口资源，同时也受限于DDR带宽。

HDMI/DP 转MIPI-CSI规格：

| 芯片型号    | MIPI RX数量                                        | 最大分辨率(PHY规格)                                                                                                                                                                                            | 支持格式(MIPI RX)                   | 逐行隔行 |
|---------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|------|
| RK3588s | 3路：<br>3x4lane<br>4路：<br>2x2lane<br>+<br>2x4lane | DPHY:<br>1xDPHY-v1.2: 2.5G/lane x 4lane<br>4K60YUV422<br>DCPHY:<br>(1)DPHY: 2xDPHY-v2.0: 2.5Gbps/lane x 4lane 4K60YUV422<br>(2)CPHY: 2xCPHY-v1.1:<br>2.5Gsp/trio(5.7Gbps) x 3trios<br>4K60YUV422/RGB888 | YUV422/RGB888/<br>RGB666/RGB565 | P/I  |
| RK3588  | 4路：<br>4x4lane<br>6路：<br>4x2lane<br>+<br>2x4lane | DPHY:<br>2xDPHY-v1.2: 2.5G/lane x 4lane<br>4K60YUV422<br>DCPHY:<br>(1)DPHY: 2xDPHY-v2.0: 2.5G/lane x 4lane<br>4K60YUV422<br>(2)CPHY: 2xCPHY-v1.1:<br>2.5Gsp/trio(5.7Gbps) x 3trios<br>4K60YUV422/RGB888 | YUV422/RGB888/<br>RGB666/RGB565 | P/I  |
| RK356X  | 1x4lane<br>or<br>2x2lane                         | 1xDPHY-v1.2: 2.5G/lane x 4lane<br>4K60YUV422                                                                                                                                                            | YUV422/RGB888/<br>RGB666/RGB565 | P/I  |
| RK3562  | 2x4lane<br>or<br>4x2lane                         | 2xDPHY-v1.2: 2.5G/lane x 4lane<br>4K60YUV422                                                                                                                                                            | YUV422/RGB888/<br>RGB666/RGB565 | P/I  |

注：CPHY描述带宽吞吐率单位为sps，与bps的换算：1 Gsp = 2.28 Gbps。

HDMI To BT1120规格受限于PCLK大小，规格如下：

| 芯片型号    | 支持数量 | 最大分辨率   | 支持格式   | 逐行隔行支持  |
|---------|------|---------|--------|---------|
| RK3588s | 1    | 双边沿4K30 | YUV422 | 支持逐行/隔行 |
| RK3588  | 1    | 双边沿4K30 | YUV422 | 支持逐行/隔行 |
| RK356X  | 1    | 双边沿4K30 | YUV422 | 支持逐行/隔行 |

注：RK3562不支持BT1120接口。

### 1.3 已验证桥接芯片介绍

常见的一些验证过的桥接芯片如下表所示：具体可以参考文档：  
《Rockchip\_HDMI\_To\_CSI\_Bridge\_Chips\_Summary\_V1.0.1.pdf》

| 转接芯片                                                         | 功能                                           | 工作模式/<br>验证最大分辨率/格式                                                | 逐<br>行/<br>隔行 | 备注                |
|--------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------|---------------|-------------------|
| IT6510<br>DPHY:2.5Gbps/lane                                  | DP to CSI                                    | DPHY: 4K30/YUV422                                                  | P             |                   |
| IT6616<br>DPHY:2Gbps/lane                                    | HDMI to CSI                                  | DPHY: 4K30/YUV422                                                  | P             |                   |
| RK628D<br>DPHY: 1.2Gbps/lane                                 | HDMI to CSI<br>HDMI to DSI<br>HDMI to BT1120 | DPHY: 4K30/YUV422<br>DSI: RGB888 1080P60<br>BT1120: 1080P60        | P             |                   |
| RK628F<br>DPHY: 1.5Gbps/lane                                 | HDMI to CSI<br>HDMI to DSI<br>HDMI to BT1120 | DPHY: 4K60/YUV422<br>DSI: RGB888 4KP60<br>BT1120: 1080P60          | P             | 4K60<br>双<br>PORT |
| LT6911C<br>DPHY:1.5Gbps/lane                                 | HDMI to CSI                                  | DPHY: 1080P60/YUV422                                               | P             |                   |
| LT6911UXC<br>DPHY: 2Gbps/lane                                | HDMI to CSI                                  | DPHY:4K30/YUV422<br>4K60/YUV422(dual mipi)                         | P             | 4K60<br>双<br>PORT |
| LT6911UXE<br>DPHY:2.5Gbps/lane                               | HDMI to CSI                                  | DPHY:4K60/YUV422<br>4K60/RGB888/dual mipi                          | P/I           |                   |
| LT7911D<br>DPHY:1.5Gbps/lane                                 | DP/Type-c to CSI                             | DPHY:4K30/YUV422<br>4K60/YUV422(dual mipi)                         | P             |                   |
| LT7911UXC<br>DPHY:2.5Gbps/lane<br>CPHY:2.5Gsps/trio(5.7Gbps) | DP/Type-c to CSI                             | DPHY:4K60/YUV422<br>DPHY:4K60/RGB888(dualmipi)<br>CPHY:5K60/YUV422 | P             |                   |
| LT8619C<br>PCLK:148.5M DDR                                   | HDMI to<br>BT1120/BT656                      | BT1120:1080P60/YUV422                                              | P/I           |                   |
| LT8668SX<br>DPHY:2.5Gbps/lane<br>CPHY:2.5Gsps/trio(5.7Gbps)  | HDMI to CSI<br>DP/type-c to CSI              | CPHY:5K60/YUV422                                                   | P             |                   |
| TC358749XBG<br>DPHY:1Gbps/lane                               | HDMI to CSI                                  | DPHY:1080P60/YUV422                                                | P/I           |                   |

### 1.4 与Android11/10/9 平台差异点

Android11/10/9平台HDMIIN功能基于CameraHAL3开发，可以参考文档  
《Rockchip\_Developer\_Guide\_HDMI\_IN\_Based\_On\_CameraHal3\_CN.pdf》。

Android12+ HDMIIN主要变化有：优化了分辨率变化与热拔插功能，增加了低延时方案TV框架的支持。

### 1.4.1 分辨率变化/热拔插功能优化

Android11/10/9平台，分辨率变化与热拔插功能主要依赖于APK轮询驱动节点，查询分辨率的变化与拔插状态，来实现分辨率切换功能和热拔插功能。

Android12+ 优化了分辨率变化与热拔插的流程，由驱动识别到分辨率变化或者拔插变化，主动通过V4L2框架上报v4l2-event事件，应用订阅事件，获取到事件后，针对事件的类型进行重新开关流的处理，保证图像恢复正常。

### 1.4.2 支持TIF框架

Android11/10/9平台都是基于CameraHAL3框架实现，将转接芯片当做类camera设备点亮。

Android12+ 增加了TVinput的支持，实现低延时的预览。

## 2. 驱动代码与关键配置

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### 2.1 驱动代码和kernel配置

#### 2.1.1 驱动代码

转接芯片主要驱动代码如下：

```
drivers/media/i2c/rk628/  
drivers/media/i2c/lt6911uxe.c  
drivers/media/i2c/lt7911uxc.c  
drivers/media/i2c/lt6911uxc.c  
drivers/media/i2c/lt7911d.c  
drivers/media/i2c/it6616.c  
drivers/media/i2c/tc35874x.c
```

vicap控制器/mipi-csi驱动代码如下：

```
drivers/media/platform/rockchip/cif/
```

#### 2.1.2 config配置

kernel的config配置如下：

```
CONFIG_VIDEO_LT6911UXC=y  
CONFIG_VIDEO_LT6911UXE=y  
CONFIG_VIDEO_LT7911D=y  
CONFIG_VIDEO_NVP6188=y  
CONFIG_VIDEO_RK628_CSI=y  
CONFIG_VIDEO_RK628_BT1120=y  
CONFIG_VIDEO_TC35874X=y
```

控制器相关的配置：

```
CONFIG_VIDEO_ROCKCHIP_CIF=y
CONFIG_PHY_ROCKCHIP_CSI2_DPHY=y
```

## 2.2 DTS配置

### 2.2.1 转接芯片dts配置

转接芯片dts的配置如下，是i2c设备，需要配置到i2c总线下：

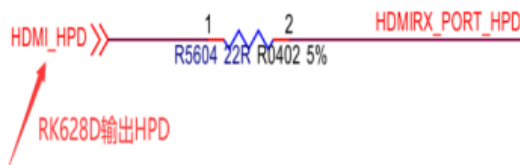
```
&i2c5 {
    status = "okay";

    rk628_csi: rk628_csi@50 {
        reg = <0x50>;
        compatible = "rockchip,rk628-csi-v4l2";
        status = "okay";
        power-domains = <&power RK3588_PD_VI>;
        pinctrl-names = "default";
        pinctrl-0 = <&rk628_pin>;
        interrupt-parent = <&gpio2>;
        interrupts = <RK_PC4 IRQ_TYPE_LEVEL_HIGH>;
        enable-gpios = <&gpio1 RK_PA0 GPIO_ACTIVE_HIGH>;
        reset-gpios = <&gpio4 RK_PC6 GPIO_ACTIVE_HIGH>;
        plugin-det-gpios = <&gpio1 RK_PA1 GPIO_ACTIVE_LOW>;
        continues-clk = <1>;

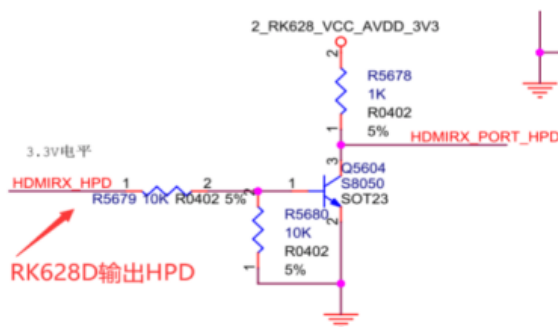
        rockchip,camera-module-index = <0>;
        rockchip,camera-module-facing = "back";
        rockchip,camera-module-name = "HDMI-MIPI";
        rockchip,camera-module-lens-name = "RK628-CSI";
        port {
            hdmiin_out0: endpoint {
                remote-endpoint = <&hdmi_mipi0_in>;
                data-lanes = <1 2 3 4>;
            };
        };
    };
};
```

- interrupt-parent/ interrupts: 连接RK628D中断的GPIO引脚；
- enable-gpios: RK628D供电控制GPIO引脚（若为常供电可不配置）；
- reset-gpios: RK628D复位控制GPIO引脚
- hpd-output-inverted: HPD输出取反配置，若HPD输出电平在电路上做了取反，则需要使能此配置项；

◆ HPD未取反设计:

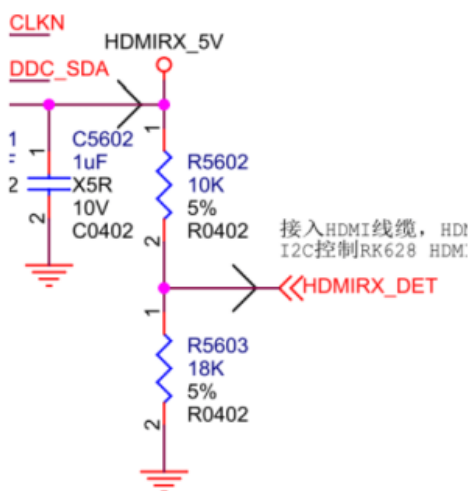


◆ HPD取反设计:

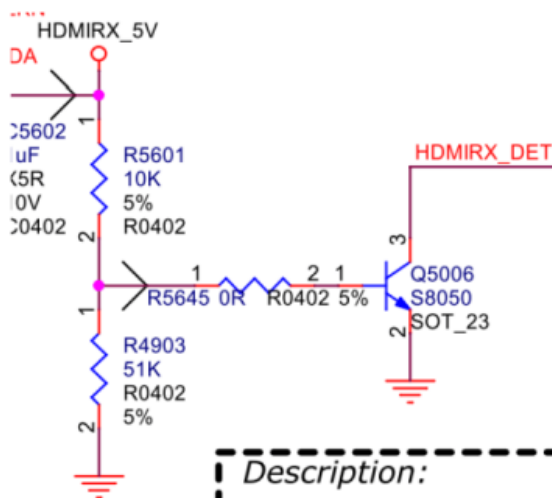


- plugin-det-gpios: HDMI插入检测GPIO引脚, 注意电路上有对电平取反, 有效电平需要配置正确;

◆ HDMIRX\_DET未取反设计:



◆ HDMIRX\_DET取反设计:



- rockchip,camera-module-name: 新版本对接TIF框架, 该属性必须严格按照如下规律配置: HDMI-MIPIx, 其中x为硬件实际接到哪个MIPI就配置哪个。主要在RK3588上面存在疑问, 按照如下规律配置。

若硬件接的是MIPI D/C PHY0, 则为HDMI-MIPI0或者HDMI-MIPI

若硬件接的是MIPI D/C PHY1, 则为HDMI-MIPI1

若硬件接的是MIPI CSI RX0 (4lane), 则为HDMI-MIPI2

若硬件接的是MIPI CSI RX0 (2lane-lane01), 则为HDMI-MIPI2

若硬件接的是MIPI CSI RX0 (2lane-lane23), 则为HDMI-MIPI3

若硬件接的是MIPI CSI RX1 (4lane), 则为HDMI-MIPI4

若硬件接的是MIPI CSI RX1 (2lane-lane01), 则为HDMI-MIPI4

若硬件接的是MIPI CSI RX1 (2lane-lane23), 则为HDMI-MIPI5

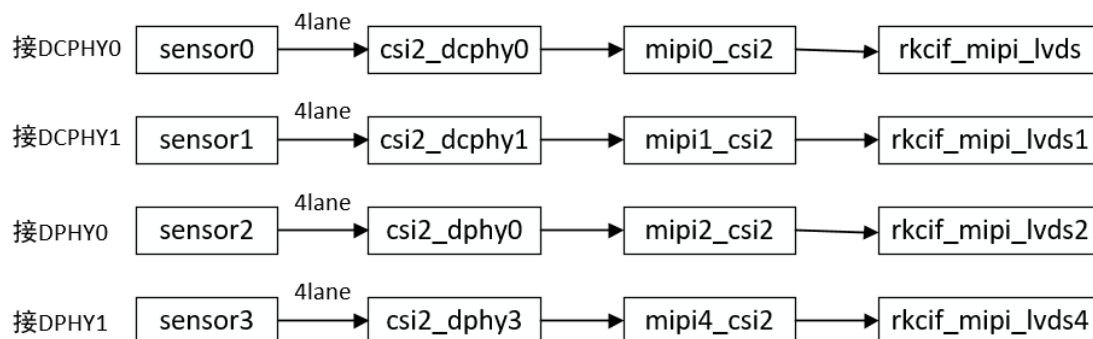
另外从DTS的pipeline链路也可以看出啊实际的配置, 该属性与rkcif\_mipi\_lvds对应, rkcif\_mipi\_lvds对接的index是多少, 则该属性也配置为多少。



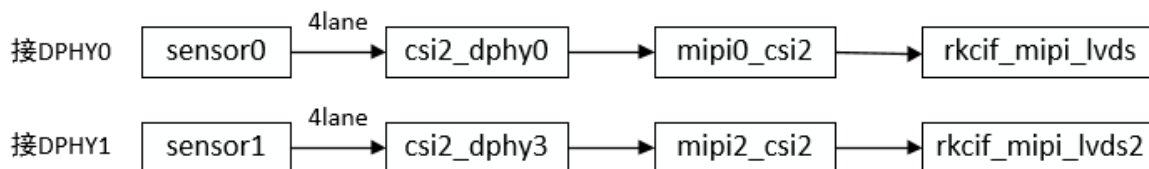
## 2.2.2 pipeline配置

图像接收链路需要根据接的硬件接口进行配置，RK356X只有一路MIPI，因此不用做区分，RK3588有4路4lane的MIPI，需要区分，RK3562有2路4lane的MIPI也需要区分。

RK3588图像链路可参考如下：



RK3562参考如下：



这里以RK628接RK3588的DCPHY0为例：

```
&csi2_dcphy0 {
    status = "okay";

    ports {
        #address-cells = <1>;
        #size-cells = <0>;
        port@0 {
            reg = <0>;
            #address-cells = <1>;
            #size-cells = <0>;

            hdmi_mipi0_in: endpoint@1 {
                reg = <1>;
                remote-endpoint = <&hdmi_in_out0>;
                data-lanes = <1 2 3 4>;
            };
        };
        port@1 {
            reg = <1>;
            #address-cells = <1>;
            #size-cells = <0>;

            csidcphy0_out: endpoint@0 {
                reg = <0>;
                remote-endpoint = <&mipi0_csi2_input>;
            };
        };
    };
};
```

```

    };

};

&mipi_dcphy0 {
    status = "okay";
};

&mipi0_csi2 {
    status = "okay";

    ports {
        #address-cells = <1>;
        #size-cells = <0>;

        port@0 {
            reg = <0>;
            #address-cells = <1>;
            #size-cells = <0>;

            mipi0_csi2_input: endpoint@1 {
                reg = <1>;
                remote-endpoint = <&csidcphy0_out>;
            };
        };

        port@1 {
            reg = <1>;
            #address-cells = <1>;
            #size-cells = <0>;

            mipi0_csi2_output: endpoint@0 {
                reg = <0>;
                remote-endpoint = <&cif_mipi_in0>;
            };
        };
    };
};

&rkCIF {
    status = "okay";
};

&rkCIF_mipi_lvds {
    status = "okay";

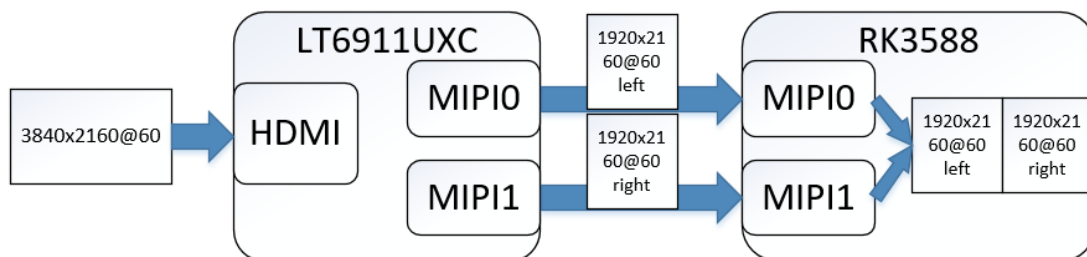
    port {
        cif_mipi_in0: endpoint {
            remote-endpoint = <&mipi0_csi2_output>;
        };
    };
};

&rkCIF_mmu {
    status = "okay";
};

```

## 2.3 双MIPI拼接方案

针对部分转接芯片，受限于MIPI DPHY带宽，4lane的场景无法支持4K60的场景（4K60 YUV422需要10Gbps带宽，4lane每条lane需要支持2.5Gbps），可以使用空余的MIPI接口来拼接支持4K60。需要转接芯片支持双MIPI输出，主控双MIPI接收进行拼接，目前RK3588、RK3562支持，系统框架如下：



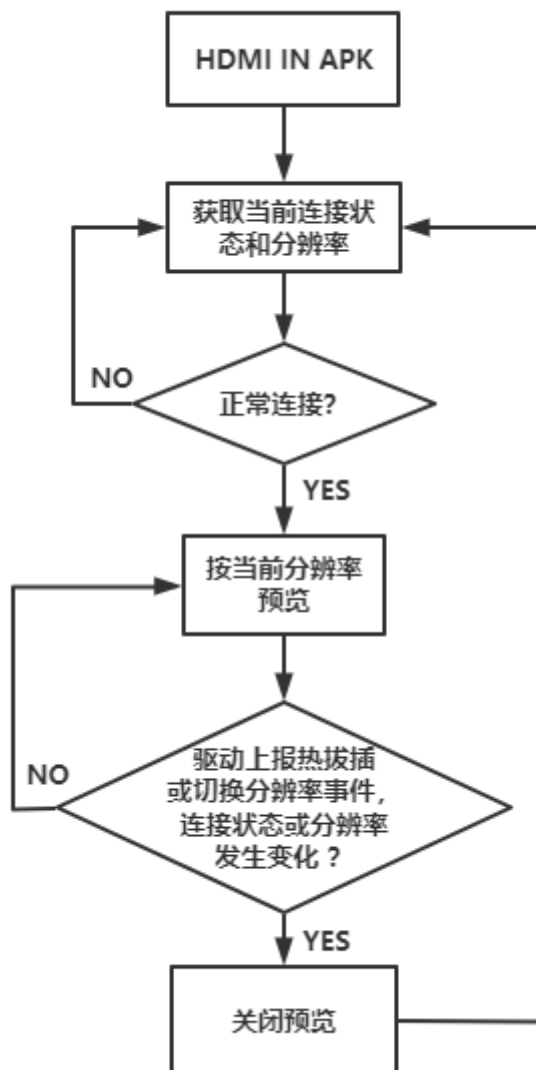
更多关于双MIPI拼接接收4K60的可以参考文档：

《Rockchip\_RK3588\_HDMI\_To\_CSI\_Dual\_Mipi\_Developer\_Guide.pdf》。

## 3. HDMI IN VIDEO框架

### 3.1 HDMI IN VIDEO工作流程

HDMI转MIPI-CSI实现HDMI IN功能流程主要如下图所示：



## 3.2 HDMI2MIPI 主要驱动架构

HDMI TO MIPI-CSI驱动框架基于V4l2框架和media-framework的架构设计实现，并增加了分辨率切换功能以及热拔插功能的设计，基于V4l2/media-framework的驱动框架设计可以参考文档：

《Rockchip\_Driver\_Guide\_VI\_CN\_v1.1.4.pdf》，这里重点关注分辨率变化与热拔插功能相关的几个中断和delayed\_work：

- plugin\_detect\_irq: 5V检测中断，用于检测HDMI接口的拔插动作。
- rk628\_csi\_irq\_handler: HDMI RX中断，监测HDMI信号变化时使用。
- delayed\_work\_hotplug: 5V检测中断对应的delayed\_work，产生热拔插动作时，进行相应的配置处理，并设置拔插状态，上报拔插事件。
- delayed\_work\_res\_change: HDMI RX控制器检测到信号变化中断时，对转接芯片进行重新配置，获取分辨率，上报分辨率变化事件。
- subscribe\_event: 注册v4l2事件，主要注册了分辨率变化事件与拔插事件。

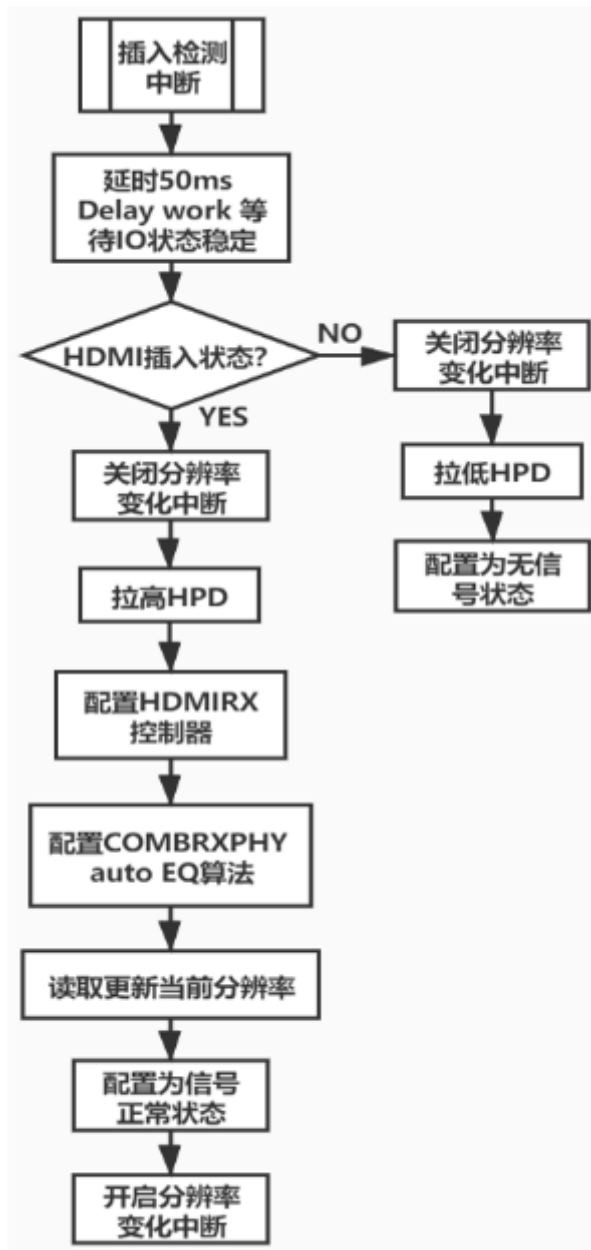
驱动主要流程参考RK628流程，主要包括：初始化、热拔插中断处理、分辨率切换中断处理。



驱动初始配置流程



分辨率变化中断处理程序



## 热拔插中断处理程序

### 3.3 图像传输延时

- 对接TIF框架后加速显示，延时约为：30~50ms。
- 对接Camera框架，显示延时约为：100~120ms

## 4. HDMIIN APK 适配方法

### 4.1 APK源码

- packages/apps/TV/partner\_support/samples：提供TV源数据服务，通过framework与HAL层、预览APK进行交互，由于是开机运行的隐藏服务，该APK在桌面上是隐藏图标。

- hardware/rockchip/tv\_input : TVHAL层代码，开关流、热拔插和分辨率切换事件等与驱动进行命令交互。
- hardware/rockchip/camera: cameraHAL层代码，camera框架取流、拍照等功能实现。
- vendor/rockchip/hardware/interfaces/hdmi: 走camera框架使用，负责监听分辨率变化与热拔插事件，与驱动以及APK交互。
- packages/apps/rkCamera2 : 预览apk，此应用有2个界面。默认的MainActivity界面使用TIF方案进行预览，通过framework层与上述TV源数据服务进行交互；RockchipCamera2界面使用Camera方案进行预览，用标准的Camera API，对MIPI的节点的cameraid进行open操作。  
APK在桌面上图标名称为 HdmiIn ，通常客户会二次开发替换为自己的预览APK。
- SDK默认代码HDMI IN功能是关闭的，使能HDMI IN功能，需配置如下属性，开启后会编译含上述APK在内的相关模块：

```
vim device/rockchip/rk3588/BoardConfig.mk
BOARD_HDMI_IN_SUPPORT := true
```

- 走camera框架需要打开：

```
vim device/rockchip/rk3588/BoardConfig.mk
CAMERA_SUPPORT_HDMI := true
```

只配置CAMERA\_SUPPORT\_HDMI，而不配置BOARD\_HDMI\_IN\_SUPPORT的情况下。如想要用rkCamera2进行camera预览，需要将rkCamera2加入到编译并配置属性persist.sys.hdmiinmode值为2。其他补充见文档：

《Rockchip\_Developer\_Guide\_HDMI\_RX\_CN》

```
PRODUCT_PACKAGES += \
    rkCamera2
```

## 4.2 camera3\_profiles.xml配置

走camera框架预览需要适配camera3\_profiles.xml，注册camera设备。

camera3\_profiles.xml文件对应SDK目录下具体芯片平台的文件：

```
hardware/rockchip/camera/etc/camera/camera3_profiles_rk3xxx.xml
```

主要配置注意事项如下，详情可参考SOC Sensor的配置方法：

- **name:** 需要与驱动名称一致，有大小写区别；
- **moduleId:** 需要与驱动dts中配置的index一致；

```
</Profiles>
<Profiles cameraId="0" name="rk628-csi" moduleId="m00">
    <Supported_hardware>
        <hwType value="SUPPORTED_HW_RKISP1"/>
    </Supported_hardware>
</Profiles>
```



```

&rk628_csi {
    status = "okay";
    /*
     * If the hpd output level is inverted on the circuit,
     * the following configuration needs to be enabled.
     */
    /* hpd-output-inverted; */
    plugin-det-gpios = <&gpio0 13 GPIO_ACTIVE_HIGH>;
    power-gpios = <&gpio0 17 GPIO_ACTIVE_HIGH>;
    rockchip,camera-module-index = <0>;
    rockchip,camera-module-facing = "back";
    rockchip,camera-module-name = "RK628-CSI";
    rockchip,camera-module-lens-name = "NC";
}

```

- scaler.availableStreamConfigurations/scaler.availableMinFrameDurations/ scaler.availableStallDurations:  
需要正确配置驱动支持的分辨率和最小的帧间隔时间，若驱动中需要增加新的分辨率支持，在这里也要相应地增加配置：

```

<scaler.availableStreamConfigurations value="BLOB,3840x2160,OUTPUT,
BLOB,1920x1080,OUTPUT,
BLOB,1280x720,OUTPUT,
BLOB,720x576,OUTPUT,
BLOB,720x480,OUTPUT,
YCbCr_420_888,3840x2160,OUTPUT,
YCbCr_420_888,1920x1080,OUTPUT,
YCbCr_420_888,1280x720,OUTPUT,
YCbCr_420_888,720x576,OUTPUT,
YCbCr_420_888,720x480,OUTPUT,
IMPLEMENTATION_DEFINED,3840x2160,OUTPUT,
IMPLEMENTATION_DEFINED,1920x1080,OUTPUT,
IMPLEMENTATION_DEFINED,1280x720,OUTPUT,
IMPLEMENTATION_DEFINED,720x576,OUTPUT,
IMPLEMENTATION_DEFINED,720x480,OUTPUT" />
<scaler.availableMinFrameDurations value="BLOB,3840x2160,33333333,
BLOB,1920x1080,16666667,
BLOB,1280x720,16666667,
BLOB,720x576,20000000,
BLOB,720x480,16666667,
YCbCr_420_888,3840x2160,33333333,
YCbCr_420_888,1920x1080,16666667,
YCbCr_420_888,1280x720,16666667,
YCbCr_420_888,720x576,20000000,
YCbCr_420_888,720x480,16666667,
IMPLEMENTATION_DEFINED,3840x2160,33333333,
IMPLEMENTATION_DEFINED,1920x1080,16666667,
IMPLEMENTATION_DEFINED,1280x720,16666667,
IMPLEMENTATION_DEFINED,720x576,20000000,
IMPLEMENTATION_DEFINED,720x480,16666667"/>
<scaler.availableStallDurations value="BLOB,3840x2160,33333333,
BLOB,1920x1080,16666667,
BLOB,1280x720,16666667,
BLOB,720x576,20000000,
BLOB,720x480,16666667"/>

```

- **sensor.orientation**: 图像旋转角度，支持0、90、180、270;

```
<sensor.maxAnalogSensitivity value="2400"/> <!-- HAL
<sensor.orientation value="0"/>
<sensor.profileHueSatMapDimensions value="0,0,0"/>
```

### 4.3 APK预览说明

APK支持RK3588 HDMI RX通路数据预览和HDMI转MIPI-CSI通路数据预览，使用时需要切换。

TIF预览方式，需要设置MIPI-CSI2通路:

```
setprop vendor.tvinput.hdmiin.type 1
```

camera预览方式，需要打开RockchipCamera2界面，注意需要使能CAMERA\_SUPPORT\_HDMI

```
setprop persist.sys.hdmiinmode 2
```

### 4.4 TIF预览与camera预览差异

|    | TIF                                                            | Camera            |
|----|----------------------------------------------------------------|-------------------|
| 优点 | 延迟低                                                            | app端能够拿到预览数据进行后处理 |
| 缺点 | 不支持屏幕旋转、分屏、画中画功能；<br>app端拿不到预览buffer数据；<br>不支持screencap方式的截图命令 | 延迟高于TIF           |

## 5. 常用调试方法

### 5.1 打开log开关

一般调试过程中建议打开debug的log开关:

打开桥接芯片debug开关:

```
echo 1 > /sys/module/rk628_csi/parameters/debug
```

打开vicap驱动debug开关:

```
echo 3 > /sys/module/video_rkcif/parameters/debug
echo 8 > /proc/sys/kernel/printk
```

打开cameraHAL3 debug开关:

```
setprop persist.vendor.camera.hal.debug 5
```

## 5.2 查询所有video节点与subdev节点

可以通过如下命令查询到转接芯片（RK628）对应的subdev节点：

```
rk3588_s:/ # grep -H '' /sys/class/video4linux/v4l-subdev*/name
/sys/class/video4linux/v4l-subdev0/name:rockchip-mipi-csi2
/sys/class/video4linux/v4l-subdev1/name:rockchip-csi2-dphy0
/sys/class/video4linux/v4l-subdev2/name:m00_b_rk628-csi 5-0050
/sys/class/video4linux/v4l-subdev3/name:rockchip-mipi-csi2
/sys/class/video4linux/v4l-subdev4/name:rockchip-csi2-dphy0
```

查询video节点：

```
rk3588_s:/ # grep -H '' /sys/class/video4linux/video*/name
/sys/class/video4linux/video0/name:stream_cif_mipi_id0
/sys/class/video4linux/video1/name:stream_cif_mipi_id1
/sys/class/video4linux/video10/name:rkcif_tools_id2
/sys/class/video4linux/video11/name:stream_cif_mipi_id0
/sys/class/video4linux/video12/name:stream_cif_mipi_id1
/sys/class/video4linux/video13/name:stream_cif_mipi_id2
/sys/class/video4linux/video14/name:stream_cif_mipi_id3
/sys/class/video4linux/video15/name:rkcif_scale_ch0
/sys/class/video4linux/video16/name:rkcif_scale_ch1
/sys/class/video4linux/video17/name:rkcif_scale_ch2
/sys/class/video4linux/video18/name:rkcif_scale_ch3
/sys/class/video4linux/video19/name:rkcif_tools_id0
/sys/class/video4linux/video2/name:stream_cif_mipi_id2
/sys/class/video4linux/video20/name:rkcif_tools_id1
/sys/class/video4linux/video21/name:rkcif_tools_id2
/sys/class/video4linux/video22/name:stream_hdmirx
/sys/class/video4linux/video3/name:stream_cif_mipi_id3
/sys/class/video4linux/video4/name:rkcif_scale_ch0
/sys/class/video4linux/video5/name:rkcif_scale_ch1
/sys/class/video4linux/video6/name:rkcif_scale_ch2
/sys/class/video4linux/video7/name:rkcif_scale_ch3
/sys/class/video4linux/video8/name:rkcif_tools_id0
/sys/class/video4linux/video9/name:rkcif_tools_id1
```

## 5.3 查询驱动timings信息

```
rk3588_s:/ # v4l2-ctl -d /dev/v4l-subdev2 --query-dv-timings
Active width: 1920
Active height: 1080
Total width: 2200
Total height: 1125
Frame format: progressive
Polarities: -vsync -hsync
Pixelclock: 0 Hz (0.00 frames per second)
Horizontal frontporch: 88
```

```
Horizontal sync: 0
Horizontal backporch: 192
Vertical frontporch: 9
Vertical sync: 0
Vertical backporch: 36
Standards:
```

## 5.4 查询分辨率和图像格式

对转接芯片所在的subdev节点查询分辨率和图像格式:

```
rk3588_s:/ # v4l2-ctl -d /dev/v4l-subdev2 --get-subdev-fmt
ioctl: VIDIOC_SUBDEV_G_FMT (pad=0)
  Width/Height      : 1920/1080
  Mediabus Code     : 0x2006
  Field             : None
  Colospace         : Default
  Transfer Function : Default (maps to Rec. 709)
  YCbCr/HSV Encoding: Default (maps to ITU-R 601)
  Quantization      : Default (maps to Limited Range)
```

## 5.5 查看media拓扑结构

HDMI2MIPI驱动框架类似camera，需要保证pipeline的完整才可以正常工作，使用media-ctl查看pipeline:

```
media-ctl -d /dev/mediaX -p //X=0123...
```

以RK3588 + RK628为例，正常结果pipeline显示如下:

```
|rk3588_s:/ # media-ctl -p
Opening media device /dev/media0
Enumerating entities
Found 14 entities
Enumerating pads and links
Media controller API version 0.0.160

Media device information
-----
driver      rkcif
model       rkcif-mipi-lvds
serial
bus info
hw revision 0x0
driver version 0.0.160

Device topology
- entity 1: stream_cif_mipi_id0 (1 pad, 11 links)
    type Node subtype V4L
    device node name /dev/video0
    pad0: Sink
        <- "rockchip-mipi-csi2":1 [ENABLED]
        <- "rockchip-mipi-csi2":2 []
```

```

        <- "rockchip-mipi-csi2":3 []
        <- "rockchip-mipi-csi2":4 []
        <- "rockchip-mipi-csi2":5 []
        <- "rockchip-mipi-csi2":6 []
        <- "rockchip-mipi-csi2":7 []
        <- "rockchip-mipi-csi2":8 []
        <- "rockchip-mipi-csi2":9 []
        <- "rockchip-mipi-csi2":10 []
        <- "rockchip-mipi-csi2":11 []

- entity 5: stream_cif_mipi_id1 (1 pad, 11 links)
  type Node subtype V4L
  device node name /dev/video1
  pad0: Sink
    <- "rockchip-mipi-csi2":1 []
    <- "rockchip-mipi-csi2":2 [ENABLED]
    <- "rockchip-mipi-csi2":3 []
    <- "rockchip-mipi-csi2":4 []
    <- "rockchip-mipi-csi2":5 []
    <- "rockchip-mipi-csi2":6 []
    <- "rockchip-mipi-csi2":7 []
    <- "rockchip-mipi-csi2":8 []
    <- "rockchip-mipi-csi2":9 []
    <- "rockchip-mipi-csi2":10 []
    <- "rockchip-mipi-csi2":11 []

- entity 9: stream_cif_mipi_id2 (1 pad, 11 links)
  type Node subtype V4L
  device node name /dev/video2
  pad0: Sink
    <- "rockchip-mipi-csi2":1 []
    <- "rockchip-mipi-csi2":2 []
    <- "rockchip-mipi-csi2":3 [ENABLED]
    <- "rockchip-mipi-csi2":4 []
    <- "rockchip-mipi-csi2":5 []
    <- "rockchip-mipi-csi2":6 []
    <- "rockchip-mipi-csi2":7 []
    <- "rockchip-mipi-csi2":8 []
    <- "rockchip-mipi-csi2":9 []
    <- "rockchip-mipi-csi2":10 []
    <- "rockchip-mipi-csi2":11 []

- entity 13: stream_cif_mipi_id3 (1 pad, 11 links)
  type Node subtype V4L
  device node name /dev/video3
  pad0: Sink
    <- "rockchip-mipi-csi2":1 []
    <- "rockchip-mipi-csi2":2 []
    <- "rockchip-mipi-csi2":3 []
    <- "rockchip-mipi-csi2":4 [ENABLED]
    <- "rockchip-mipi-csi2":5 []
    <- "rockchip-mipi-csi2":6 []
    <- "rockchip-mipi-csi2":7 []
    <- "rockchip-mipi-csi2":8 []
    <- "rockchip-mipi-csi2":9 []
    <- "rockchip-mipi-csi2":10 []
    <- "rockchip-mipi-csi2":11 []

```

```

- entity 17: rkCIF_scale_ch0 (1 pad, 11 links)
    type Node subtype V4L
    device node name /dev/video4
    pad0: Sink
        <- "rockchip-mipi-csi2":1 []
        <- "rockchip-mipi-csi2":2 []
        <- "rockchip-mipi-csi2":3 []
        <- "rockchip-mipi-csi2":4 []
        <- "rockchip-mipi-csi2":5 [ENABLED]
        <- "rockchip-mipi-csi2":6 []
        <- "rockchip-mipi-csi2":7 []
        <- "rockchip-mipi-csi2":8 []
        <- "rockchip-mipi-csi2":9 []
        <- "rockchip-mipi-csi2":10 []
        <- "rockchip-mipi-csi2":11 []

- entity 21: rkCIF_scale_ch1 (1 pad, 11 links)
    type Node subtype V4L
    device node name /dev/video5
    pad0: Sink
        <- "rockchip-mipi-csi2":1 []
        <- "rockchip-mipi-csi2":2 []
        <- "rockchip-mipi-csi2":3 []
        <- "rockchip-mipi-csi2":4 []
        <- "rockchip-mipi-csi2":5 []
        <- "rockchip-mipi-csi2":6 [ENABLED]
        <- "rockchip-mipi-csi2":7 []
        <- "rockchip-mipi-csi2":8 []
        <- "rockchip-mipi-csi2":9 []
        <- "rockchip-mipi-csi2":10 []
        <- "rockchip-mipi-csi2":11 []

- entity 25: rkCIF_scale_ch2 (1 pad, 11 links)
    type Node subtype V4L
    device node name /dev/video6
    pad0: Sink
        <- "rockchip-mipi-csi2":1 []
        <- "rockchip-mipi-csi2":2 []
        <- "rockchip-mipi-csi2":3 []
        <- "rockchip-mipi-csi2":4 []
        <- "rockchip-mipi-csi2":5 []
        <- "rockchip-mipi-csi2":6 []
        <- "rockchip-mipi-csi2":7 [ENABLED]
        <- "rockchip-mipi-csi2":8 []
        <- "rockchip-mipi-csi2":9 []
        <- "rockchip-mipi-csi2":10 []
        <- "rockchip-mipi-csi2":11 []

- entity 29: rkCIF_scale_ch3 (1 pad, 11 links)
    type Node subtype V4L
    device node name /dev/video7
    pad0: Sink
        <- "rockchip-mipi-csi2":1 []
        <- "rockchip-mipi-csi2":2 []
        <- "rockchip-mipi-csi2":3 []
        <- "rockchip-mipi-csi2":4 []
        <- "rockchip-mipi-csi2":5 []
        <- "rockchip-mipi-csi2":6 []

```

```

        <- "rockchip-mipi-csi2":7 []
        <- "rockchip-mipi-csi2":8 [ENABLED]
        <- "rockchip-mipi-csi2":9 []
        <- "rockchip-mipi-csi2":10 []
        <- "rockchip-mipi-csi2":11 []

- entity 33: rkCIF_tools_id0 (1 pad, 11 links)
  type Node subtype V4L
  device node name /dev/video8
  pad0: Sink
    <- "rockchip-mipi-csi2":1 []
    <- "rockchip-mipi-csi2":2 []
    <- "rockchip-mipi-csi2":3 []
    <- "rockchip-mipi-csi2":4 []
    <- "rockchip-mipi-csi2":5 []
    <- "rockchip-mipi-csi2":6 []
    <- "rockchip-mipi-csi2":7 []
    <- "rockchip-mipi-csi2":8 []
    <- "rockchip-mipi-csi2":9 [ENABLED]
    <- "rockchip-mipi-csi2":10 []
    <- "rockchip-mipi-csi2":11 []

- entity 37: rkCIF_tools_id1 (1 pad, 11 links)
  type Node subtype V4L
  device node name /dev/video9
  pad0: Sink
    <- "rockchip-mipi-csi2":1 []
    <- "rockchip-mipi-csi2":2 []
    <- "rockchip-mipi-csi2":3 []
    <- "rockchip-mipi-csi2":4 []
    <- "rockchip-mipi-csi2":5 []
    <- "rockchip-mipi-csi2":6 []
    <- "rockchip-mipi-csi2":7 []
    <- "rockchip-mipi-csi2":8 []
    <- "rockchip-mipi-csi2":9 []
    <- "rockchip-mipi-csi2":10 [ENABLED]
    <- "rockchip-mipi-csi2":11 []

- entity 41: rkCIF_tools_id2 (1 pad, 11 links)
  type Node subtype V4L
  device node name /dev/video10
  pad0: Sink
    <- "rockchip-mipi-csi2":1 []
    <- "rockchip-mipi-csi2":2 []
    <- "rockchip-mipi-csi2":3 []
    <- "rockchip-mipi-csi2":4 []
    <- "rockchip-mipi-csi2":5 []
    <- "rockchip-mipi-csi2":6 []
    <- "rockchip-mipi-csi2":7 []
    <- "rockchip-mipi-csi2":8 []
    <- "rockchip-mipi-csi2":9 []
    <- "rockchip-mipi-csi2":10 []
    <- "rockchip-mipi-csi2":11 [ENABLED]

- entity 45: rockchip-mipi-csi2 (12 pads, 122 links)
  type V4L2 subdev subtype Unknown
  device node name /dev/v4l-subdev0
  pad0: Sink

```

```
[fmt:UYVY2X8/1920x1080
crop.bounds:(0,0)/1920x1080
crop:(0,0)/1920x1080]
<- "rockchip-csi2-dphy0":1 [ENABLED]
pad1: Source
-> "stream_cif_mipi_id0":0 [ENABLED]
-> "stream_cif_mipi_id1":0 []
-> "stream_cif_mipi_id2":0 []
-> "stream_cif_mipi_id3":0 []
-> "rkcif_scale_ch0":0 []
-> "rkcif_scale_ch1":0 []
-> "rkcif_scale_ch2":0 []
-> "rkcif_scale_ch3":0 []
-> "rkcif_tools_id0":0 []
-> "rkcif_tools_id1":0 []
-> "rkcif_tools_id2":0 []
pad2: Source
-> "stream_cif_mipi_id0":0 []
-> "stream_cif_mipi_id1":0 [ENABLED]
-> "stream_cif_mipi_id2":0 []
-> "stream_cif_mipi_id3":0 []
-> "rkcif_scale_ch0":0 []
-> "rkcif_scale_ch1":0 []
-> "rkcif_scale_ch2":0 []
-> "rkcif_scale_ch3":0 []
-> "rkcif_tools_id0":0 []
-> "rkcif_tools_id1":0 []
-> "rkcif_tools_id2":0 []
pad3: Source
-> "stream_cif_mipi_id0":0 []
-> "stream_cif_mipi_id1":0 []
-> "stream_cif_mipi_id2":0 [ENABLED]
-> "stream_cif_mipi_id3":0 []
-> "rkcif_scale_ch0":0 []
-> "rkcif_scale_ch1":0 []
-> "rkcif_scale_ch2":0 []
-> "rkcif_scale_ch3":0 []
-> "rkcif_tools_id0":0 []
-> "rkcif_tools_id1":0 []
-> "rkcif_tools_id2":0 []
pad4: Source
-> "stream_cif_mipi_id0":0 []
-> "stream_cif_mipi_id1":0 []
-> "stream_cif_mipi_id2":0 []
-> "stream_cif_mipi_id3":0 [ENABLED]
-> "rkcif_scale_ch0":0 []
-> "rkcif_scale_ch1":0 []
-> "rkcif_scale_ch2":0 []
-> "rkcif_scale_ch3":0 []
-> "rkcif_tools_id0":0 []
-> "rkcif_tools_id1":0 []
-> "rkcif_tools_id2":0 []
pad5: Source
-> "stream_cif_mipi_id0":0 []
-> "stream_cif_mipi_id1":0 []
-> "stream_cif_mipi_id2":0 []
-> "stream_cif_mipi_id3":0 []
-> "rkcif_scale_ch0":0 [ENABLED]
```



```

-> "rkCIF_scale_ch1":0 []
-> "rkCIF_scale_ch2":0 []
-> "rkCIF_scale_ch3":0 []
-> "rkCIF_tools_id0":0 []
-> "rkCIF_tools_id1":0 []
-> "rkCIF_tools_id2":0 []
pad6: Source
-> "stream_cif_mipi_id0":0 []
-> "stream_cif_mipi_id1":0 []
-> "stream_cif_mipi_id2":0 []
-> "stream_cif_mipi_id3":0 []
-> "rkCIF_scale_ch0":0 []
-> "rkCIF_scale_ch1":0 [ENABLED]
-> "rkCIF_scale_ch2":0 []
-> "rkCIF_scale_ch3":0 []
-> "rkCIF_tools_id0":0 []
-> "rkCIF_tools_id1":0 []
-> "rkCIF_tools_id2":0 []
pad7: Source
-> "stream_cif_mipi_id0":0 []
-> "stream_cif_mipi_id1":0 []
-> "stream_cif_mipi_id2":0 []
-> "stream_cif_mipi_id3":0 []
-> "rkCIF_scale_ch0":0 []
-> "rkCIF_scale_ch1":0 []
-> "rkCIF_scale_ch2":0 [ENABLED]
-> "rkCIF_scale_ch3":0 []
-> "rkCIF_tools_id0":0 []
-> "rkCIF_tools_id1":0 []
-> "rkCIF_tools_id2":0 []
pad8: Source
-> "stream_cif_mipi_id0":0 []
-> "stream_cif_mipi_id1":0 []
-> "stream_cif_mipi_id2":0 []
-> "stream_cif_mipi_id3":0 []
-> "rkCIF_scale_ch0":0 []
-> "rkCIF_scale_ch1":0 []
-> "rkCIF_scale_ch2":0 []
-> "rkCIF_scale_ch3":0 [ENABLED]
-> "rkCIF_tools_id0":0 []
-> "rkCIF_tools_id1":0 []
-> "rkCIF_tools_id2":0 []
pad9: Source
-> "stream_cif_mipi_id0":0 []
-> "stream_cif_mipi_id1":0 []
-> "stream_cif_mipi_id2":0 []
-> "stream_cif_mipi_id3":0 []
-> "rkCIF_scale_ch0":0 []
-> "rkCIF_scale_ch1":0 []
-> "rkCIF_scale_ch2":0 []
-> "rkCIF_scale_ch3":0 []
-> "rkCIF_tools_id0":0 [ENABLED]
-> "rkCIF_tools_id1":0 []
-> "rkCIF_tools_id2":0 []
pad10: Source
-> "stream_cif_mipi_id0":0 []
-> "stream_cif_mipi_id1":0 []
-> "stream_cif_mipi_id2":0 []

```

```

-> "stream_cif_mipi_id3":0 []
-> "rkcif_scale_ch0":0 []
-> "rkcif_scale_ch1":0 []
-> "rkcif_scale_ch2":0 []
-> "rkcif_scale_ch3":0 []
-> "rkcif_tools_id0":0 []
-> "rkcif_tools_id1":0 [ENABLED]
-> "rkcif_tools_id2":0 []

pad11: Source
-> "stream_cif_mipi_id0":0 []
-> "stream_cif_mipi_id1":0 []
-> "stream_cif_mipi_id2":0 []
-> "stream_cif_mipi_id3":0 []
-> "rkcif_scale_ch0":0 []
-> "rkcif_scale_ch1":0 []
-> "rkcif_scale_ch2":0 []
-> "rkcif_scale_ch3":0 []
-> "rkcif_tools_id0":0 []
-> "rkcif_tools_id1":0 []
-> "rkcif_tools_id2":0 [ENABLED]

- entity 58: rockchip-csi2-dphy0 (2 pads, 2 links)
    type V4L2 subdev subtype Unknown
    device node name /dev/v4l-subdev1
    pad0: Sink
        [fmt:UYVY2X8/1920x1080]
        <- "m00_b_rk628-csi 5-0050":0 [ENABLED]
    pad1: Source
        -> "rockchip-mipi-csi2":0 [ENABLED]

- entity 63: m00_b_rk628-csi 5-0050 (1 pad, 1 link)
    type V4L2 subdev subtype Sensor
    device node name /dev/v4l-subdev2
    pad0: Source
        [fmt:UYVY2X8/1920x1080]
        -> "rockchip-csi2-dphy0":0 [ENABLED]

```

## 5.6 开启图像数据流

开启图像数据流命令如下所示，需要根据实际情况，设置video节点、分辨率、格式等信息，video节点可以按照如下方法确认：使用上述查看media拓扑命令找到转接芯片所在的media节点并查看对应的pipeline，pipeline中节点名为：stream\_cif\_mipi\_id0所对应的video节点既是取流的节点：

```

v4l2-ctl --verbose -d /dev/video0 --set-fmt-
video=width=3840,height=2160,pixelformat='NV12' --stream-mmap=4

```

## 5.7 抓取图像文件

可以将图像保存到文件，通过adb pull 到设备，使用7yuv，YUVview等工具软件查看：

```
v4l2-ctl --verbose -d /dev/video0 --set-fmt-  
video=width=3840,height=2160,pixelformat='NV12' --stream-mmap=3 --stream-skip=4 -  
-stream-to=/data/3840x2160_nv12.yuv --stream-count=5 --stream-poll
```

## 5.8 正常取流log

若一切正常，能正常接收到图像，执行完上述抓数据流的命令，应会正常打印帧率，如出现卡住没帧率刷新或者帧率不对等问题，都需要进行排查。

```
rk3588_s:/ # v4l2-ctl --verbose -d /dev/video0 --set-fmt-  
video=width=1920,height=1080,pixelformat='NV12' --stream-mmap=4  
VIDIOC_QUERYCAP: ok  
VIDIOC_G_FMT: ok  
VIDIOC_S_FMT: ok  
Format Video Capture Multiplanar:  
    Width/Height      : 1920/1080  
    Pixel Format       : 'NV12'  
    Field              : None  
    Number of planes  : 1  
    Flags              :  
    Colospace         : Default  
    Transfer Function : Default  
    YCbCr Encoding    : Default  
    Quantization      : Default  
    Plane 0           :  
        Bytes per Line : 1920  
        Size Image      : 3110400  
VIDIOC_REQBUFS: ok  
VIDIOC_QUERYBUF: ok  
VIDIOC_QUERYBUF: ok  
VIDIOC_QBUF: ok  
VIDIOC_QUERYBUF: ok  
VIDIOC_QBUF: ok  
VIDIOC_QUERYBUF: ok  
VIDIOC_QBUF: ok  
VIDIOC_QUERYBUF: ok  
VIDIOC_QBUF: ok  
VIDIOC_STREAMON: ok  
idx: 0 seq:      0 bytesused: 3110400 ts: 249.466825  
idx: 1 seq:      1 bytesused: 3110400 ts: 249.483367 delta: 16.542 ms  
idx: 2 seq:      2 bytesused: 3110400 ts: 249.500033 delta: 16.666 ms  
idx: 3 seq:      3 bytesused: 3110400 ts: 249.516700 delta: 16.667 ms  
idx: 0 seq:      4 bytesused: 3110400 ts: 249.533367 delta: 16.667 ms fps: 60.11  
idx: 1 seq:      5 bytesused: 3110400 ts: 249.550034 delta: 16.667 ms fps: 60.09  
idx: 2 seq:      6 bytesused: 3110400 ts: 249.566700 delta: 16.666 ms fps: 60.08  
idx: 3 seq:      7 bytesused: 3110400 ts: 249.583367 delta: 16.667 ms fps: 60.06  
idx: 0 seq:      8 bytesused: 3110400 ts: 249.600034 delta: 16.667 ms fps: 60.06  
idx: 1 seq:      9 bytesused: 3110400 ts: 249.616701 delta: 16.667 ms fps: 60.05
```

## 5.9 查看camera是否注册成功

如下命令可以查看camera是否注册成功，如果没注册上，请检查camera3\_profiles.xml文件。

```
dumpsys media.camera
```

## 5.10 打开IO命令

需要在rockchip\_defconfig打开如下配置：

```
CONFIG_DEVMEM=y
```

另外还需要修改：

```
vim mkcombinedroot/configs/android-11.config
```

删除如下：

```
# CONFIG_DEVMEM is not set
```

如果需要完整编译安卓还需要修改其他依赖，参考文档

《RK3588\_Android12\_SDK\_Developer\_Guide\_CN.pdf》

# 6. 常见问题

## 6.1 抓图异常分析

### 6.1.1 抓不到图且串口没有报错

执行抓图命令，出现select timeout，说明没有收下数据，可以从以下点进行排查：

- dts接口是否配置错误，RK3588支持多路MIPI，检查硬件连接的MIPI口与dts配置的是否一致。
- 确认主控MIPIRX是否收到数据，可以直接通过硬件进行测量转接芯片是否输出，且主控端也有MIPI信号，也可以通过读取主控的io寄存器来判断DPHY的状态。

下面以RK3588为例，说明寄存器地址，可以在执行V4l2取数据流的场景下，连续读取如下寄存器(15次)，查看stopstate状态是否有变化，如果对应的bit有01变化，说明硬件上存在MIPI信号，如果没有变化则说明硬件上没有收到信号，需要检查接口配置是否正确或者硬件的连通性。

```
io -4 -1 0x100 0xfdd10000 //DCPHY0
io -4 -1 0x100 0xfdd20000 //DCPHY1
io -4 -1 0x100 0xfdd30000 //DPHY0
io -4 -1 0x100 0xfdd50000 //DPHY1
```

对应的寄存器解释如下：

## CSI2HOST\_PHY\_STATE

Address: Operational Base + offset (0x0014)

| Bit   | Attr | Reset Value | Description                                                                                                                |
|-------|------|-------------|----------------------------------------------------------------------------------------------------------------------------|
| 31:12 | RO   | 0x00000     | reserved                                                                                                                   |
| 11    | RW   | 0x0         | bypass_2ecc_tst<br>Payload Bypass test mode for double ECC errors.                                                         |
| 10    | RO   | 0x0         | phy_stopstateclk<br>Clock lane in Stop state.                                                                              |
| 9     | RO   | 0x0         | phy_rxulpsclknot<br>This signal indicates that the clock lane module has entered the Ultra Low Power state.<br>Active low. |
| 8     | RO   | 0x0         | phy_rxclkactivehs<br>Indicates that the clock lane is actively receiving a DDR clock.                                      |
| 7     | RO   | 0x0         | phy_stopstatedata_3<br>Data lane 3 in Stop state.                                                                          |
| 6     | RO   | 0x0         | phy_stopstatedata_2<br>Data lane 2 in Stop state.                                                                          |
| 5     | RO   | 0x0         | phy_stopstatedata_1<br>Data lane 1 in Stop state.                                                                          |
| 4     | RO   | 0x0         | phy_stopstatedata_0<br>Data lane 0 in Stop state.                                                                          |
| 3     | RO   | 0x0         | phy_rxulpsesc_3<br>Lane module0 has entered the Ultra Low Power mode.                                                      |

如果读取的stopstate状态正常在01之间变化，但是执行V4I2抓数据没有反应且串口也没有报错，那么较大概率是转接芯片TX的数据流早于主控的数据流。

- MIPI信号是连续模式还是非连续模式，如是连续模式，检查是否是转接芯片数据流默认打开，一般要求先打开主控的接收，再打开数据流，建议在非连续模式下调试，非连续模式没有这个要求。

### 6.1.2 抓不到数据MIPI相关报错

抓不到数据，比较常见的是出现MIPI相关的报错，CRC ECC等错误，可以排查以下问题：

- 驱动代码 link freq的配置，该配置为mipi clk，即速率的一半，需要与TX端的MIPI 速率对应配置

如果没有条件测试转接芯片的输出MIPI CLK，那么可以按照如下方式进行计算：

MIPI CLK = Htotal \* Vtotal \* 16 \* fps / 4 / 2，例如1080P60根据公式计算如下：

$2200 * 1125 * 16 * 60 / 4 / 2 = 297M$ 。

- dts MIPI lane数是否配置错误，驱动接口g\_mbus接口lane数是否配置错误，需要根据实际情况配置。
- MIPI信号质量问题，检查是否存在飞线，走线过长等问题
- TX端的MIPI信号问题，检查MIPI信号是否符合SPEC的指标要求
- 如果接的是RK3588的DCPHY，检查MIPI的LP11时序是否符合DCPHY的设计要求。

### 6.1.3 抓图SIZE ERROR报错

需要排查设置的抓图的分辨率大小是否超过实际输入源的大小。VICAP控制器不带放大的功能，因此出现这个报错时可以先检查一下是否满足

OUTPUT\_SIZE <= INPUT\_SIZE

## 6.1.4 OPEN 节点错误

节点出现错误通常会出现如下问题：

- 打不开节点No such device报错

需要检查链路是否正常，链路不完整会导致这个问题；链路不完整包括是否有相关的驱动节点没注册上，包括：转接芯片、csi\_dphy、mipi-csi、stream\_cif等节点；链路是否成功enable，如果先打开了apk，有可能会清空链路导致这个问题。

- 出现Invalid arguments的报错

检查抓图的命令是否正确，参数是否正确，逗号空格等是否正确使用。

- 出现device busy等报错

检查节点是否被占用。

## 6.2 media链路异常

media链路是否正常可以通过工具进行查看，确认每一级的节点都正常，才可以正常使用，这个case的链路配置如下：rk628 -> csi\_dphy -> mipi-csi2 -> rkCIF\_mipi\_lvds

不需要使能对应的rkCIF\_mipi\_lvds\_stdif节点，也不需要进入ISP，因此不需要使能ISP的节点。如果将链路配置到了ISP，从ISP获取数据流会导致获取不到数据流。

## 6.3 1080P图像预览黑屏

走TV框架的时候，可能会出现4K分辨率预览正常，1080P无法预览现象。logcat如下，出现qbuf失败：

```
rk3588_s:/ # logcat | grep tv_input
12-11 07:02:56.561  483  483 D tv_input_HinDevImpl: [acquire_buffer 1334]
VIDIOC_QUERYBUF successful
12-11 07:02:56.561  483  483 E tv_input_HinDevImpl: start_device:line 696 |
VIDIOC_QBUF Failed, error: Invalid argument
12-11 07:02:56.561  483  483 E tv_input_HinDevImpl: start:line 747 | Start v4l2
device failed:-1
```

此时可以打开vicap驱动的debug开关，看内核log打印：

```
echo 3 > /sys/module/video_rkcif/parameters/debug
echo 8 > /proc/sys/kernel/printk
```

内核log如下：

```
[ 7013.981832][ T483] rkCIF-mipi-lvds2: remote fmt: mbus code:0x2006,
size:1920x1080, field: 1
[ 7013.981907][ T483] rkCIF-mipi-lvds2: C-Plane 0 size: 2073600, Total
imagesize: 2073600
[ 7013.981919][ T483] rkCIF-mipi-lvds2: C-Plane 1 size: 2073600, Total
imagesize: 4147200
[ 7013.981933][ T483] rkCIF-mipi-lvds2: rkCIF_set_fmt: req(1920, 1080) out(1920,
1080)
[ 7013.982665][ T483] rkCIF-mipi-lvds2: vid-cap-mplane count 5, size 4177920,
extended(0, 0)
```

上述log可以看到应用申请的buf大小是4147200，驱动计算的需要大小4177920，申请的buf大小小于驱动计算的，导致失败，原因是vicap驱动默认高16位对齐，所以1080P的分辨率会按照1920x1088x2来计算。可增加如下修改，取消驱动高16对齐的限制：

```
diff --git a/drivers/media/platform/rockchip/cif/capture.c
b/drivers/media/platform/rockchip/cif/capture.c
index 457a1f755b1af..47b0e9b5f2dcf 100644
--- a/drivers/media/platform/rockchip/cif/capture.c
+++ b/drivers/media/platform/rockchip/cif/capture.c
@@ -4280,7 +4280,8 @@ static int rkCIF_queue_setup(struct vb2_queue *queue,

    for (i = 0; i < cif_fmt->mplanes; i++) {
        const struct v4l2_plane_pix_format *plane_fmt;
-       int h = round_up(height, MEMORY_ALIGN_ROUND_UP_HEIGHT);
+       int h = dev->chip_id >= CHIP_RK3588_CIF ?
+           height : round_up(height, MEMORY_ALIGN_ROUND_UP_HEIGHT);

        plane_fmt = &pixm->plane_fmt[i];
        sizes[i] = plane_fmt->sizeimage / height * h;
```

## 6.4 出现reqbufs空指针异常

出现类似如下log的空指针异常，是因为转接芯片驱动上报给控制器的是0x0的分辨率，导致申请buf的异常，需要检查一下分辨率是否正常，是否插入源等等。

```
[27030.881249][ Cam3ReqThread-19257 C3....] rkCIF-mipi-lvds2: remote fmt: mbus
code:0x2006, size:0x0, field: 1
[27030.881262][ Cam3ReqThread-19257 C3....] rkCIF-mipi-lvds2: C-Plane 0 size: 0,
Total imagesize: 0
[27030.881271][ Cam3ReqThread-19257 C3....] rkCIF-mipi-lvds2: C-Plane 1 size: 0,
Total imagesize: 0
[27030.881280][ Cam3ReqThread-19257 C3....] rkCIF-mipi-lvds2: rkCIF_set_fmt:
req(0, 0) out(0, 0)
[27030.881385][ Cam3ReqThread-19257 C3....] rkCIF-mipi-lvds2: vid-cap-mplane
count 4, size 0, extended(0, 0)
[27030.881445][ Cam3ReqThread-19257 C3d..2] -----[ cut here ]-----
[27030.881464][ Cam3ReqThread-19257 C3d..2] WARNING: CPU: 3 PID: 19257 at
drivers/media/common/videobuf2/videobuf2-core.c:806 vb2_core_reqbufs+0x3b8/0x3e4
[27030.881473][ Cam3ReqThread-19257 C3d..2] Modules linked in: r8168
[27030.881493][ Cam3ReqThread-19257 C3d..2] CPU: 3 PID: 19257 Comm:
Cam3ReqThread Tainted: G          W          5.10.110 #146
```

```

[27030.881502][ Cam3ReqThread-19257 C3d..2] Hardware name: Rockchip RK3588 NVR
DEMO LP4 V10 Android Board (DT)
[27030.881513][ Cam3ReqThread-19257 C3d..2] pstate: 60400009 (nZCv daif +PAN -
UAO -TCO BTYPE=--)
[27030.881523][ Cam3ReqThread-19257 C3d..2] pc : vb2_core_reqbufs+0x3b8/0x3e4
[27030.881533][ Cam3ReqThread-19257 C3d..2] lr : vb2_core_reqbufs+0x230/0x3e4
[27030.881541][ Cam3ReqThread-19257 C3d..2] sp : ffffffff018a0ba20
[27030.881549][ Cam3ReqThread-19257 C3d..2] x29: ffffffff018a0ba50 x28:
0000000000000008
[27030.881562][ Cam3ReqThread-19257 C3d..2] x27: ffffffff018a0bbd0 x26:
ffffffffff0118c0c40
[27030.881575][ Cam3ReqThread-19257 C3d..2] x25: ffffffff81aa19fc00 x24:
ffffffffff81a0e32900
[27030.881588][ Cam3ReqThread-19257 C3d..2] x23: 0000000000000000 x22:
ffffffffff8106f90658
[27030.881600][ Cam3ReqThread-19257 C3d..2] x21: 0000000000000001 x20:
ffffffffff018a0bbd0
[27030.881613][ Cam3ReqThread-19257 C3d..2] x19: ffffffff8106f905f0 x18:
ffffffffff0139f5070
[27030.881625][ Cam3ReqThread-19257 C3d..2] x17: 0000000000000000 x16:
0000000000000090
[27030.881638][ Cam3ReqThread-19257 C3d..2] x15: 0000000000000004 x14:
00000000000003968
[27030.881650][ Cam3ReqThread-19257 C3d..2] x13: ffffffff012132a90 x12:
0000000000000003

```

## 6.5 关数据流空指针异常，导致重启

报错log类似如下，在关流的时候导致重启：

```

[ 88.101018][ T499] rkCIF-mipi-lvds2: stream[0] start stopping, total mode
0x1, cur 0x1
[ 88.101029][ T499] rkCIF-mipi-lvds2: get vblank fail, vblank_def 0,
vblank_curr 0
[ 88.101068][ T499] rockchip-mipi-csi2 fdd30000.mipi2-csi2: stream off,
src_sd: 00000000c6e16906, sd_name:rockchip-csi2-dphy0
[ 88.101072][ T499] rockchip-mipi-csi2 fdd30000.mipi2-csi2: stream OFF
[ 88.102104][ T499] rockchip-csi2-dphy csi2-dphy0: csi2_dphy_s_stream_stop
stream stop, dphy0
[ 88.102111][ T499] rockchip-csi2-dphy csi2-dphy0: csi2_dphy_s_stream stream
on:0, dphy0
[ 88.102125][ T499] rockchip-csi2-dphy csi2-dphy0: csi2_dphy_s_stream stream
on:0, dphy0
[ 88.102129][ T499] m00_b_LT6911UXC 8-002b: enable_stream: disable
[ 88.102145][ T499] Unable to handle kernel NULL pointer dereference at
virtual address 0000000000000028
[ 88.230007][ T499] Mem abort info:
[ 88.233589][ T499]   ESR = 0x96000005
[ 88.237396][ T499]   EC = 0x25: DABT (current EL), IL = 32 bits
[ 88.243444][ T499]   SET = 0, FnV = 0
[ 88.247258][ T499]   EA = 0, S1PTW = 0
[ 88.251125][ T499] Data abort info:
[ 88.254744][ T499]   ISV = 0, ISS = 0x00000005
[ 88.259332][ T499]   CM = 0, WnR = 0
[ 88.263021][ T499] user pgtable: 4k pages, 39-bit VAs, pgdp=0000000127a2e000

```



```
[ 88.270322][ T499] [0000000000000028] pgd=0000000000000000,
p4d=0000000000000000, pud=0000000000000000
[ 88.279755][ T499] Internal error: Oops: 96000005 [#1] PREEMPT SMP
[ 88.286022][ T499] Modules linked in: bcmhdhd dhd_static_buf r8168
[ 88.292196][ T499] CPU: 4 PID: 499 Comm: input@1.0-servi Not tainted 5.10.107
#50
[ 88.299751][ T499] Hardware name: Rockchip RK3588 EVB1 LP4 V10 Board (DT)
[ 88.299753][ T499] pstate: a0400009 (NzCv daif +PAN -UAO -TCO BTYPE=--)
[ 88.299759][ T499] pc : media_entity_remote_pad+0x1c/0x50
[ 88.299762][ T499] lr : rkCIF_do_stop_stream+0x584/0xa24
[ 88.299767][ T499] sp : ffffffff017f1ba30
[ 88.313834][ T499] x29: ffffffff017f1baa0 x28: ffffffff8103351028
[ 88.313836][ T499] x27: ffffffff81519ec3b0 x26: ffffffff81519ee3b0
[ 88.313838][ T499] x25: 0000000000000000 x24: ffffffff8103351058
[ 88.326012][ T499] x23: ffffffff81033500bc x22: ffffffff8103351038
[ 88.326014][ T499] x21: ffffffff81033518d2 x20: ffffffff8103350600
[ 88.326015][ T499] x19: ffffffff8103350080 x18: ffffffff01349e078
[ 88.326017][ T499] x17: ffffffff017f1b771 x16: ffffffff0000000000
[ 88.326018][ T499] x15: 0000000000000000 x14: 0000000000000008
[ 88.341835][ T499] x13: ffffffff0000000000 x12: ffffffff0121c4068
[ 88.341837][ T499] x11: 0000000000000001 x10: 0000000000000000
[ 88.341838][ T499] x9 : ffffffff8103360938 x8 : ffffffff8103360a28
[ 88.341843][ T499] x7 : 735f656c62616e65
[ 88.375645][ T499] x6 : ffffffff011f9a0a8
[ 88.375646][ T499] x5 : ffffffff017f1b928 x4 : 000000000000000f
[ 88.375648][ T499] x3 : ffffffff8103351028 x2 : ffffffff8103351028
[ 88.375649][ T499] x1 : 0000000000000000 x0 : ffffffff8103360a28
[ 88.375650][ T499] Call trace:
[ 88.375653][ T499] media_entity_remote_pad+0x1c/0x50
[ 88.375655][ T499] rkCIF_stop_streaming+0x1c/0x2c
[ 88.375660][ T499] __vb2_queue_cancel+0x3c/0x27c
[ 88.387652][ T499] vb2_core_streamoff+0x2c/0xb4
[ 88.387654][ T499] vb2_ioctl_streamoff+0x60/0x94
[ 88.387657][ T499] v4l_streamoff+0x28/0x38
[ 88.387658][ T499] __video_do_ioctl+0x2dc/0x3f8
[ 88.387663][ T499] video_usercopy+0x398/0x738
[ 88.399659][ T499] video_ioctl2+0x1c/0x2c
[ 88.399661][ T499] v4l2_ioctl+0x50/0x64
[ 88.399664][ T499] __arm64_sys_ioctl+0x94/0xd0
[ 88.399667][ T499] el0_svc_common+0xc0/0x23c
[ 88.399669][ T499] do_el0_svc+0x28/0x88
[ 88.399674][ T499] el0_svc+0x14/0x24
[ 88.411059][ T499] el0_sync_handler+0x88/0xec
[ 88.411061][ T499] el0_sync+0x1a8/0x1c0
[ 88.411062][ T499] PC: 0xfffffff0010cd7118:
[ 88.411063][ T499] 6f18 f9401108 b9406908
[ 88.423060][ T499] 34000068 128001f5 17ffffea f9400016 f9413ec8 b4000148
[ 88.423064][ T499] 6f38 f9400108 b4000108 aa1403e0 2a1303e1 2a1f03e2
d63f0100 37f80640 f9401a81
[ 88.423068][ T499] 6f58 f9401020 f9403008 b4000148 f9400508 b4000108
f9401e82 2a1303e3 d63f0100
[ 88.615992][ T499] 6f78 36f80080 2a0003f5 31080c1f 54000381 f9401e81
f9401020 f9403008 b4000288
[ 88.624848][ T499] 6f98 f9400508 b4000248 f9401a82 2a1303e3 d63f0100
36f801c0 2a0003f5 31080c1f
[ 88.633711][ T499] 6fb8 54000160 f9401a81 f9401020 f9403008 b4000168
f9400508 b4000128 f9401e82
```

```
[ 88.642567][ T499] 6fd8 b9404a83 d63f0100 14000005 f9402288 2a1f03f5
f9002697 f9002517 f9413ec8
```

关流时出现上述的报错，是链路配置异常导致的，链路配置：

rk628 -> csi\_dphy -> mipi-csi2 -> rkcif\_mipi\_lvds

不能使能rkcif\_mipi\_lvds\_stdif，该节点是连接到ISP的，单独使能该节点，却没有链接到ISP，就会导致上述问题。

## 6.6 常见log

### 6.6.1 驱动正常注册成功log

以RK628为例，驱动正常注册成功的log

```
rk3588_s:/ # dmesg | grep rk628
[ 2.088162] rk628-csi-v4l2 3-0050: RK628 I2C driver version: 00.00.08
[ 2.132529] rk628_i2c_read: i2c err reg=0x120, val=0x0 ret=-5
[ 2.132536] rk628-csi-v4l2 3-0050: RK628 MCU + I2C Mode
[ 2.132684] rk628_i2c_read: i2c err reg=0x40008, val=0x0 ret=-5
[ 2.133060] rk628-csi-v4l2 5-0050: RK628 I2C driver version: 00.00.08
[ 2.178244] rk628-csi-v4l2 5-0050: RK628 MCU + I2C Mode
[ 2.595614] m00_b_rk628-csi 5-0050: rk628-csi-v4l2 found @ 0xa0 (rk3x-i2c)
[ 2.775398] rockchip-csi2-dphy csi2-dcphy0: dphy0 matches m00_b_rk628-csi 5-0050:bus type 5
```

### 6.6.2 分辨率变化的log

以RK628为例：

```
[ 235.522728] m00_b_rk628-csi 5-0050: rk628_csi_format_change: New format:
1920x1080p60.00 (2200x1125)
[ 250.477060] rk628-csi-v4l2 5-0050: src 1920x1080 clock:148500000
[ 250.477074] rk628-csi-v4l2 5-0050: dst 1920x1080 clock:148500000
[ 250.719130] rk628-csi-v4l2 5-0050: rk628_is_avi_ready PDEC_AVI_PB:0x1010a850,
avi_rcv_rdy:1
[ 250.754242] rk628-csi-v4l2 5-0050: rk628_is_avi_ready PDEC_AVI_PB:0x1010a850,
avi_rcv_rdy:1
[ 250.790722] rk628-csi-v4l2 5-0050: rk628_is_avi_ready PDEC_AVI_PB:0x1010a850,
avi_rcv_rdy:1
[ 295.437439] m00_b_rk628-csi 5-0050: rk628_csi_format_change: New format:
1920x1080p0.00 (2200x1125)
```

## 6.7 如何配置RGB888输入

RK3588/RK3562正常是接收YUV422的图像数据，如果HDMI的输入源是RGB格式，转接芯片LT6911/RK628等CSC模块会将其转化成YUV422的CSI信号输出给到主控。RGB转YUV422的过程，存在上下采样的过程，会伴随着部分像素的丢失，颜色质量可能没有那么好，因此在部分对图像质量要求高的场景需要，需要bypass源端过来的RGB信号，RK3588/RK3562等平台支持接受CSI RGB888或者DSI

RGB888的图像信号，需要设置如下：

如果是CSI的RGB888信号，只需要在驱动设置图像格式为RGB888即可：

```
#define LT6911UXE_MEDIA_BUS_FMT          MEDIA_BUS_FMT_BGR888_1X24
```

如果转接芯片输出的是DSI的RGB888信号，除了上述的格式之外还需要设置ioctl通知vicap控制是否是DSI：

```
ioctl:
    case RKMODULE_GET_CSI_DSI_INFO:
        if (testdemo->dsi_en)
            *(int *)arg = RKMODULE_DSI_INPUT;
        else
            *(int *)arg = RKMODULE_CSI_INPUT;
        break;
ioctl32:
    case RKMODULE_GET_CSI_DSI_INFO:
        seq = kzalloc(sizeof(*seq), GFP_KERNEL);
        if (!seq) {
            ret = -ENOMEM;
            return ret;
        }

        ret = testdemo_ioctl(sd, cmd, seq);
        if (!ret) {
            ret = copy_to_user(up, seq, sizeof(*seq));
            if (ret)
                ret = -EFAULT;
        }
        kfree(seq);
        break;
```

DSI模式默认是video mode，速率高的时候建议使用command mode，转接芯片需要修改为cmd mode，主控也需要修改成command mode，在如下位置修改：

```
static int rkCIF_csi_channel_init(struct rkCIF_stream *stream,
                                struct csi_channel_info *channel)
{
    .....
    channel->cmd_mode_en = 0; /* default use DSI Video Mode */
    channel->dsi_input = dev->terminal_sensor.dsi_input_en;
    .....
}
```